Table 3: Component ablation results showing the impact of different design choices. Lower FPR95 and higher AUROC indicate better performance.

Configuration	AUROC	FPR95
ELCM (Full Method)	**91.89%**	**29.75%**
ELCM w/o Global Score	76.56%	80.50%
ELCM w/o Top-K Selection	91.45%	32.50%
ELCM w/o Spatial Correlation	91.89%	29.75%
ELCM w/ Top-3 Averaging	91.78%	29.25%

that hard selection of the most informative patches is crucial for suppressing noise from marginally relevant regions, even after entropy filtering.

Spatial Correlation Effects. Interestingly, disabling spatial correlation produces identical performance to the full method, suggesting that the entropy-based filtering and class-conditional scaling already capture most of the relevant spatial structure. This indicates that these two components are the primary drivers of our method's improvements.

This statement is incorrect. In the original code, Spatial Correlation is not applied; therefore, ELCM w/o Spatial Correlation is identical to the Full Method.

Class Pooling Strategy. Replacing max-class pooling with top-3 class averaging yields slightly lower performance (AUROC: 91.78%, FPR95: 29.25%), indicating that focusing on the single most confident class prediction per patch is more effective than averaging across multiple classes.

6.3 Alternative Scoring Functions

We evaluate alternative formulations for class-conditional weighting, finding that both ratio-based and margin-based approaches achieve similar separation quality, with ratio-based showing marginally better performance on fine-grained tasks. Both approaches create clear separation for iNaturalist while facing similar challenges with texture datasets. These comparative analyses are included in the appendix.

This subsection is written arbitrarily despite the absence of experimental results.

6.4 Component Interaction Analysis and Key Insights

Our ablation studies reveal key insights: (1) Global-local fusion is essential – the dramatic performance drop when removing global features (AUROC: 76.56%) demonstrates that local refinements complement rather than replace global semantic understanding; (2) Entropy filtering and class-conditional scaling are the primary drivers of improvement, with their combined effect significantly exceeding individual components; (3) Top-K selection provides meaningful improvements over using all filtered patches (91.45% vs 91.89% AUROC).

This was already demonstrated in GL-MCM and is not a main insight of this paper.

The stability across β values demonstrates robustness, while diminishing returns at higher values suggest moderate amplification is optimal. ELCM's improvements stem primarily from the intelligent combination of entropy-based uncertainty estimation and class-conditional discrimination enhancement, making it both effective and computationally efficient.

7 Conclusion

We presented Entropy-Weighted Local Concept Matching (ELCM), addressing limitations in existing local feature aggregation through entropy-based patch filtering, class-conditional scaling, and top-K selection with percentile-based weight stabilization.

Our method achieves overall AUROC of 91.9% and FPR95 of 29.8% compared to GL-MCM's 91.3% AUROC and 35.0% FPR95. The 5.2 percentage point FPR95 reduction represents substantial improvement in deployment reliability, with ablation studies confirming entropy filtering and class-conditional scaling as primary drivers.

The method demonstrates particular effectiveness on fine-grained recognition tasks (97.5% AUROC on iNaturalist) while providing meaningful improvements even on challenging texture-based OOD detection. Our analysis of score distributions provides insights into the method's behavior, confirming that ELCM successfully creates clearer separation between in-distribution and out-of-distribution samples across different dataset types.

Theoretical Contributions. Our work demonstrates that entropy-guided patch selection provides principled uncertainty-aware weighting, with entropy filtering and class-conditional scaling synergistically combining uncertainty estimation with discriminative amplification.

There is no theoretical contribution.

Limitations. Texture-based OOD detection remains challenging as repetitive patterns can trigger spurious local alignments despite class-conditional scaling. Primary computational overhead comes from entropy computation and top-K selection.

Concluding Remarks. ELCM represents a principled advancement in zero-shot OOD detection through intelligent local feature aggregation, establishing that entropy-guided patch selection can significantly improve upon naive pooling strategies while maintaining computational efficiency.

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